

YUKTHI 2026

NATIONAL-LEVEL HACKATHON

DATA SPECIFICATION

Intelligent Energy & Equipment Monitoring

Participant Reference Document

1. Purpose of this Document

This document defines the data contract for the YUKTHI 2026 Intelligent Energy & Equipment Monitoring challenge. It describes the structure, fields, units, temporal characteristics, missing-value conventions, and interpretation of the participant dataset.

Teams should use this specification when designing their data ingestion, preprocessing, feature engineering, machine-learning, and application pipelines. Solutions should be designed to operate on datasets that conform to this specification rather than relying on hard-coded observations.

2. Dataset Overview

The participant development dataset is provided as a CSV file containing historical observations from multiple chiller units. Each record represents an observation for a specific equipment unit at a particular timestamp.

- File format: CSV
- Development dataset size: 25,003 rows
- Number of fields: 11
- Equipment units represented in the development dataset: 3
- Equipment identifiers: CHILLER-01, CHILLER-02, CHILLER-03
- Observation period: 2019-08-18 00:00:00 to 2020-06-01 13:00:00
- Nominal observation interval: 30 minutes; occasional larger gaps are present in the data.

The development dataset does not contain an anomaly/fault target column. Teams are expected to formulate their ML approach using the information available in the dataset.

3. Data Structure

3.1 Record Identity

Each observation is uniquely identified by the combination of timestamp and equipment_id. The same timestamp may legitimately occur for different equipment units because the units are monitored independently.

Teams should therefore treat each equipment_id as its own chronological equipment series when performing time-series analysis. An identical timestamp across different equipment units is not, by itself, a duplicate record.

The supplied development dataset contains 0 duplicate (equipment_id, timestamp) pairs.

3.2 Field Specification

Field	Data Type	Unit	Description	Participant Interpretation
timestamp	Datetime	—	Observation timestamp	Use as the temporal reference for each observation. Preserve chronological ordering within each equipment unit.
equipment_id	Categorical / identifier	—	Chiller unit identifier	Identifies the equipment associated with the observation. Values in the development dataset are CHILLER-01, CHILLER-02, and CHILLER-03.
Chilled Water Rate (L/sec)	Numeric	L/sec	Chilled-water flow/rate associated with the chiller observation	Operational variable that may help characterize equipment behaviour and its relationship with load and energy consumption.
Cooling Water Temperature (C)	Numeric	°C	Cooling-water temperature	Thermal operating/context variable relevant to chiller behaviour.
Building Load (RT)	Numeric	RT	Building cooling load	Represents the cooling demand/load associated with the observation.

Chiller Energy Consumption (kWh)	Numeric	kWh	Chiller energy consumption	Primary energy-related measurement available for analysing energy behaviour.
Outside Temperature (F)	Numeric	°F	Outdoor/ambient temperature	Environmental variable that may influence equipment operating conditions.
Dew Point (F)	Numeric	°F	Outdoor dew-point temperature	Environmental/contextual variable related to atmospheric moisture conditions.
Humidity (%)	Numeric	%	Relative humidity	Environmental/contextual variable that may influence operating conditions.
Wind Speed (mph)	Numeric	mph	Outdoor wind speed	Environmental/contextual variable.
Pressure (in)	Numeric	in	Atmospheric pressure	Environmental/contextual variable.

4. Data Types and Missing Values

Numeric measurement fields are supplied as numeric values, while timestamp and equipment_id are non-numeric fields. Missing values are represented as blank/NaN values in the CSV.

Missing values are present in the development dataset. Teams should determine an appropriate strategy for handling missing data based on the characteristics of the variable and their chosen methodology.

Field	Missing Values	Share of Dataset
Chilled Water Rate (L/sec)	40	0.160%
Cooling Water Temperature (C)	5	0.020%
Building Load (RT)	19	0.076%
Chiller Energy Consumption (kWh)	9	0.036%
Humidity (%)	20	0.080%
Wind Speed (mph)	23	0.092%
Pressure (in)	12	0.048%

Fields with no missing values in the development dataset are timestamp, equipment_id, Outside Temperature (F), and Dew Point (F). This describes the supplied development data and should not be interpreted as a guarantee that any dataset provided during the challenge that conforms to this specification will have identical missingness.

5. Temporal Characteristics

The dataset is time-indexed and is intended for chronological analysis. The nominal sampling interval is 30 minutes.

- Do not assume that every equipment series is perfectly continuous.
- Gaps longer than the nominal interval exist and should be considered during temporal feature engineering and sequence-based analysis.
- Do not automatically interpret a gap as an equipment fault or anomaly; it is a data characteristic that should be handled appropriately.
- Temporal features, rolling statistics, lagged variables, persistence, and trends may be considered where appropriate to the chosen methodology.

6. Equipment-Level Analysis

The dataset contains observations for three chiller units. Equipment identifiers should be retained throughout the analysis pipeline.

- Treat equipment_id as an equipment identifier rather than a continuous numeric variable.
- Analyse temporal behaviour within each equipment unit where appropriate.
- Equipment-specific operating characteristics may be considered when establishing expected behaviour.

- Comparisons across equipment may be useful, but a difference between units should not automatically be interpreted as an anomaly.

7. Contextual and Multivariate Interpretation

The challenge is intended to support contextual anomaly detection. Participants should consider the relationship between equipment behaviour and relevant operating or environmental variables rather than relying only on isolated measurements.

Examples of contextual information available in the dataset include building load, chilled-water rate, cooling-water temperature, and outdoor environmental conditions. Teams are free to determine which variables and relationships are useful for their chosen methodology.

A measurement that appears unusual in isolation may be consistent with expected behaviour under a particular operating condition. Conversely, a value within a conventional range may still be meaningful when considered together with other variables and temporal behaviour.

8. Data Quality Considerations

Participants are responsible for determining how to prepare the supplied data for their solution. Appropriate considerations may include:

- Missing-value handling
- Outlier and noise handling
- Timestamp ordering and temporal gaps
- Equipment-specific behaviour
- Feature scaling or transformation where required
- Lagged, rolling, or historical features where appropriate
- Prevention of information leakage when validating time-dependent models

The Data Specification defines the available data; it does not prescribe a particular preprocessing or modelling method.

9. Target and Ground-Truth Information

The participant development dataset does not expose an anomaly, fault, health, or maintenance target field. Teams should not expect a labelled target column in the supplied CSV.

The absence of a target column does not prescribe a specific ML formulation. Participants may choose an appropriate methodology for learning expected behaviour and identifying deviations.

Evaluation-specific reference information is not part of the participant dataset and is addressed separately in the Evaluation Guidelines.

10. Reusable Data Pipeline Requirement

Solutions should be implemented against the data contract defined in this document. A robust implementation should be able to ingest, validate, preprocess, and analyse a dataset that follows the same field definitions and structural conventions.

- Do not hard-code individual observation values or specific timestamps.
- Do not make the application dependent on the exact number of rows in the development dataset.
- Use `equipment_id` and `timestamp` programmatically rather than relying on fixed positions or manually selected records.
- Keep data ingestion and preprocessing sufficiently modular that a conforming dataset can be processed without redesigning the application.

11. What the Data Does and Does Not Specify

The dataset provides	The dataset does not prescribe
Historical equipment observations	A specific ML algorithm
Equipment identifiers and timestamps	A specific anomaly-detection method
Energy-related measurements	A fixed feature-engineering strategy
Operational/chiller measurements	A fixed threshold or scoring formula
Environmental/contextual measurements	A specific application architecture

12. Participant Data Contract — Summary

For implementation purposes, participants should treat the following as the minimum data contract:

- CSV input with the 11 fields defined in Section 3.2.
- timestamp is the temporal reference for each observation.
- equipment_id identifies the equipment unit.
- Measurement fields are numeric and use the units specified in the field table.
- Missing values may occur in measurement fields and must be handled by the solution.
- The data is time-indexed with a nominal 30-minute interval, but gaps may occur.
- The solution should not depend on hard-coded observations or a fixed row count.
- The participant dataset contains no exposed anomaly/fault ground-truth column.

13. Relationship to the Problem Statement

This document should be read together with the YUKTHI 2026 Problem Statement. The Problem Statement defines the challenge and expected outcome; this document defines the data available to participants. Evaluation and judging procedures are documented separately in the Evaluation Guidelines.